



**CHANDIGARH
UNIVERSITY**

Discover. Learn. Empower.

Chandigarh University Uttar Pradesh

School of Computer Science & Engineering

Semester III

Python Programming

Course Code: 25 CSH-214

Faculty Name: Prateek Singhal

Python

Course Objectives

Students learn basic syntax, data types, control structures, functions, and the use of standard libraries such as NumPy and math

Students will be able to solve real-time problem-solving.

By the end of the course, students develop the ability to write efficient programs, evaluate their correctness and performance.



CO-PO Mapping

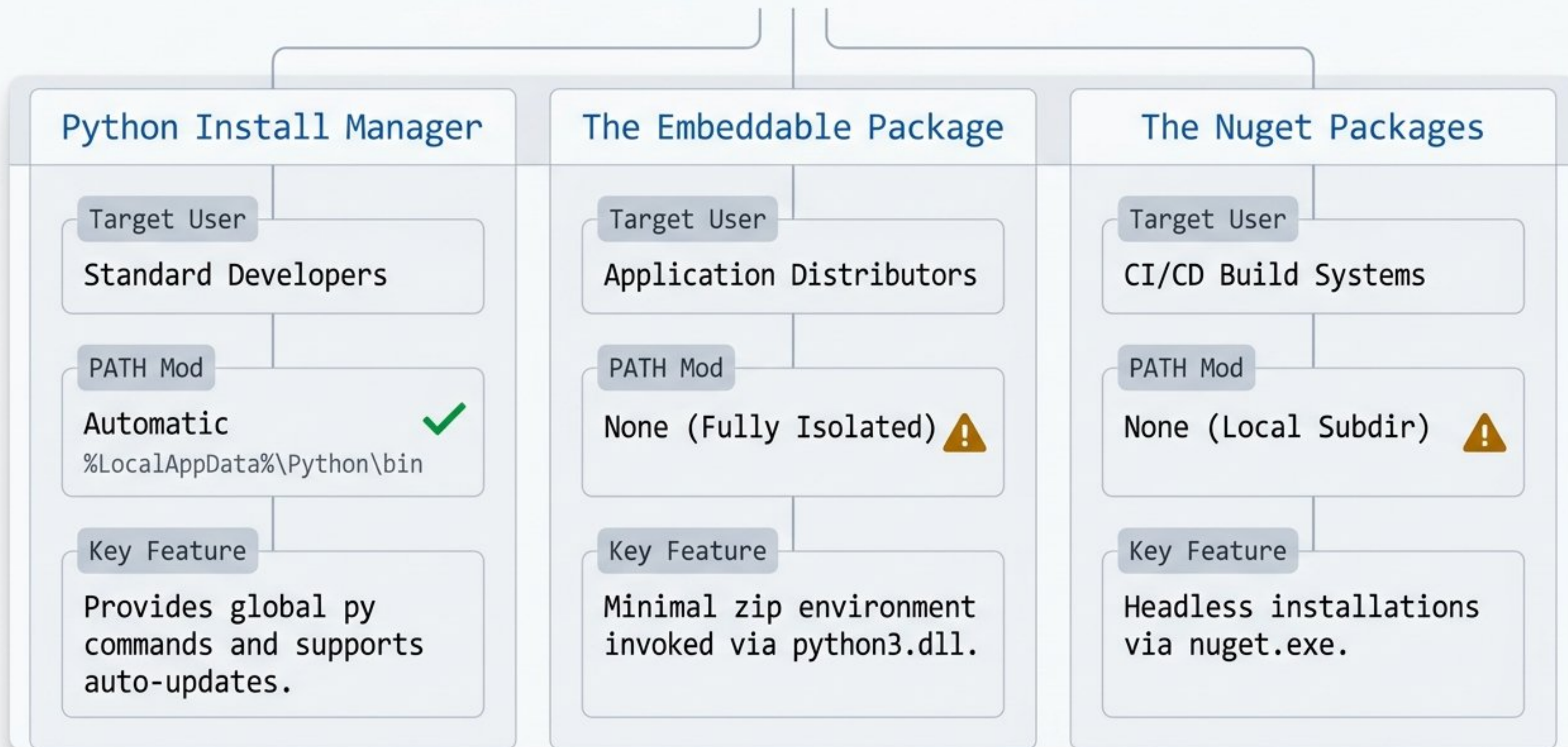
COs/ Pos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2	-	-	-	-	-	-	2	3	2
CO2	3	3	2	2	3	-	-	-	-	-	-	2	3	3
CO3	3	3	2	2	3	-	-	-	-	-	-	2	3	3
CO4	3	3	2	2	3	-	-	-	-	-	-	2	3	3
CO5	3	3	3	3	2	3	2	2	3	2	2	3	3	3

UNIT 1

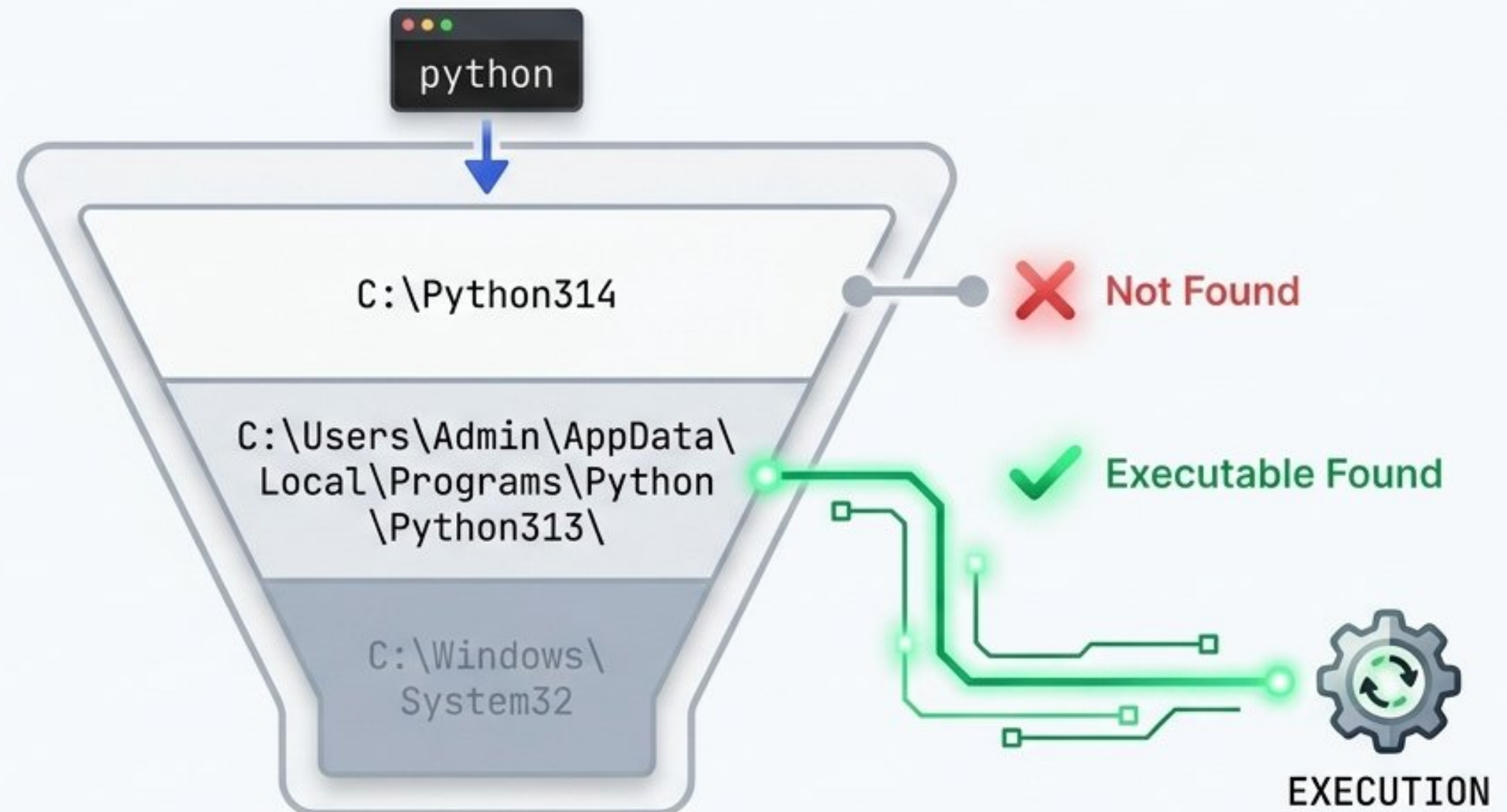
Topics to be Covered

- Python Prompt Traceback
- Handling Traceback

The Windows Installation Matrix



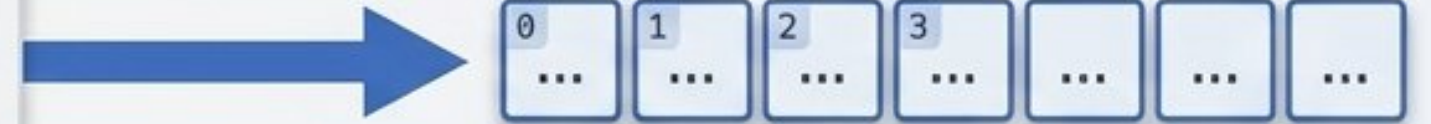
System Resolution: How Windows Finds Python



Why PATH order matters: Windows scans directories top-to-bottom. If an older Python version sits higher in the PATH variable, your newly installed version will **never execute**.

The Comprehension Cheatsheet

```
[expression for item in collection if test]
```



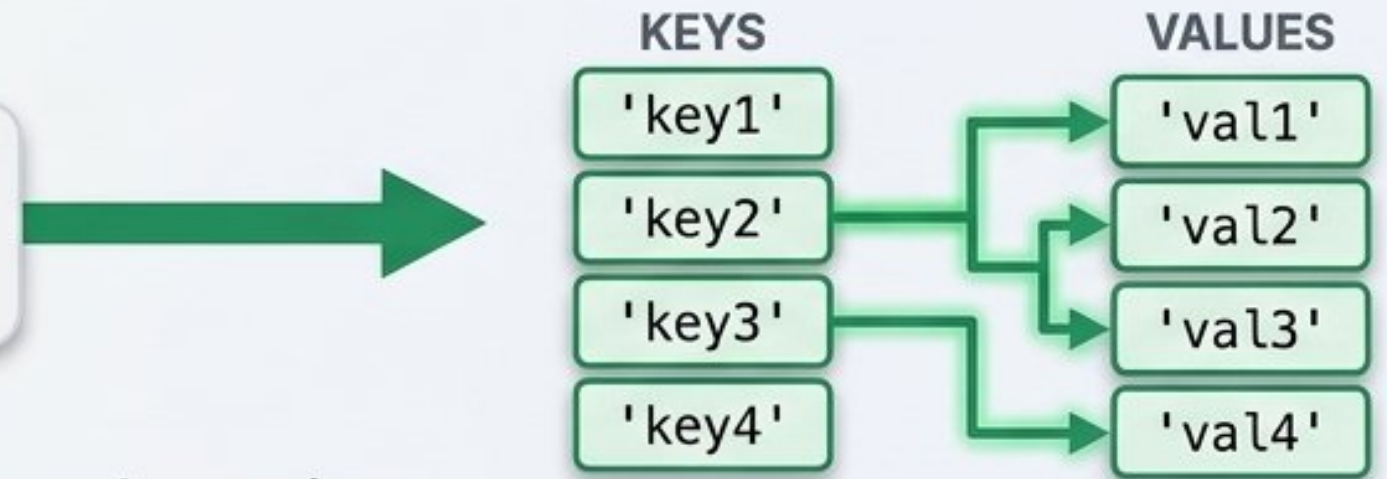
Drops elements sequentially into an ordered, indexed array.

```
{expression for item in collection}
```



Results in an unordered collection of entirely unique elements.

```
{key: value for key, value in collection}
```



Maps structured key-value pairs.

An elegant, single-line alternative to bulky *for* loops.

Generators vs. Iterables

Lists

`return list`



Computes all items at once and loads them into memory.
High risk of `MemoryError` on large datasets.

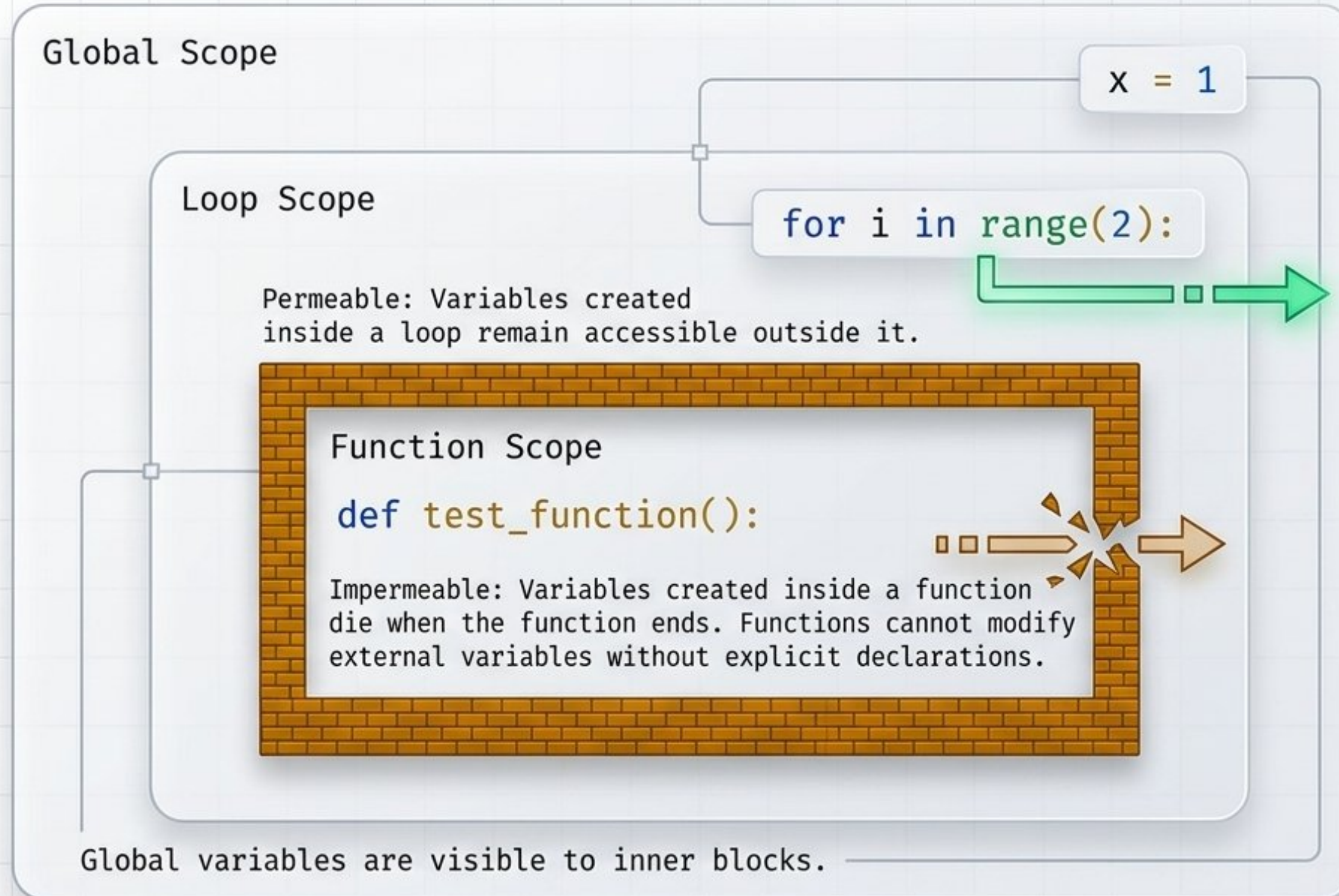
Generators

`yield n`

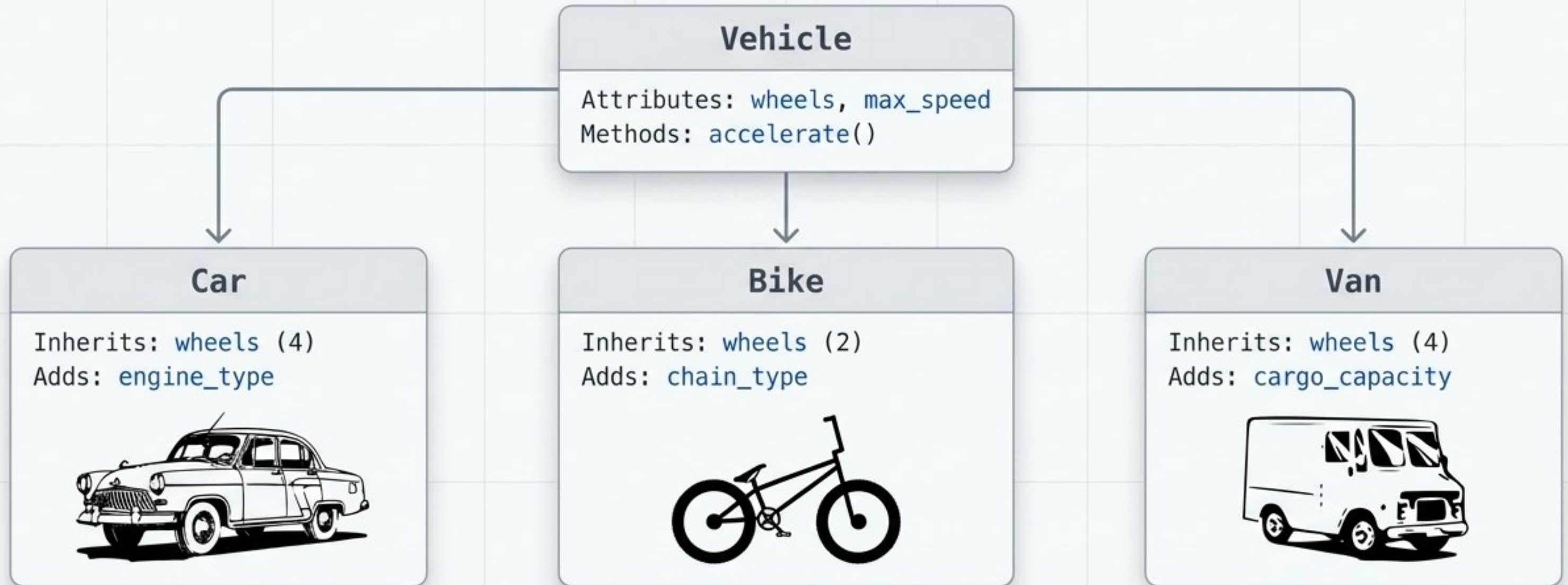


Keeps track of state and pauses execution. Consumes minimal memory regardless of dataset size.

Scope Permeability: Where Variables Live

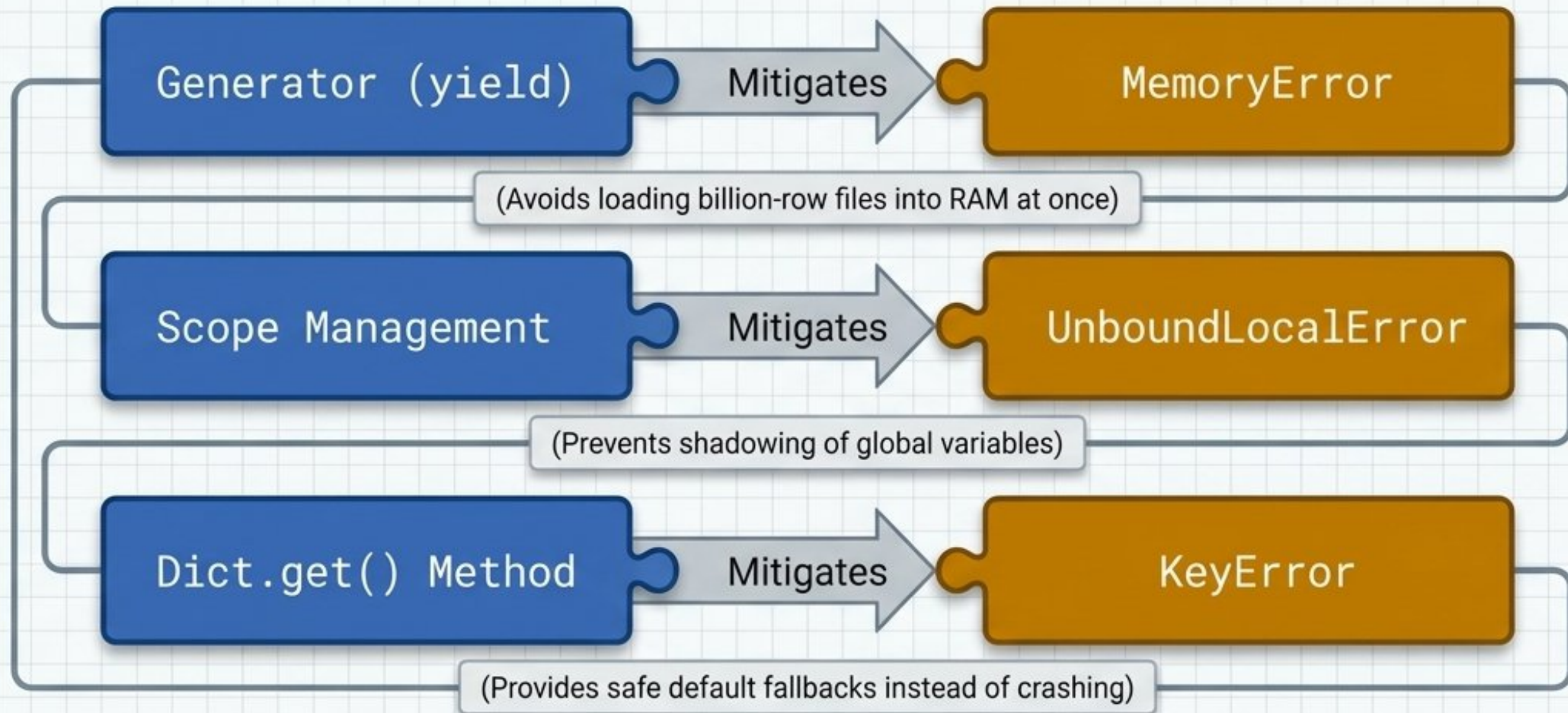


Object-Oriented Architecture



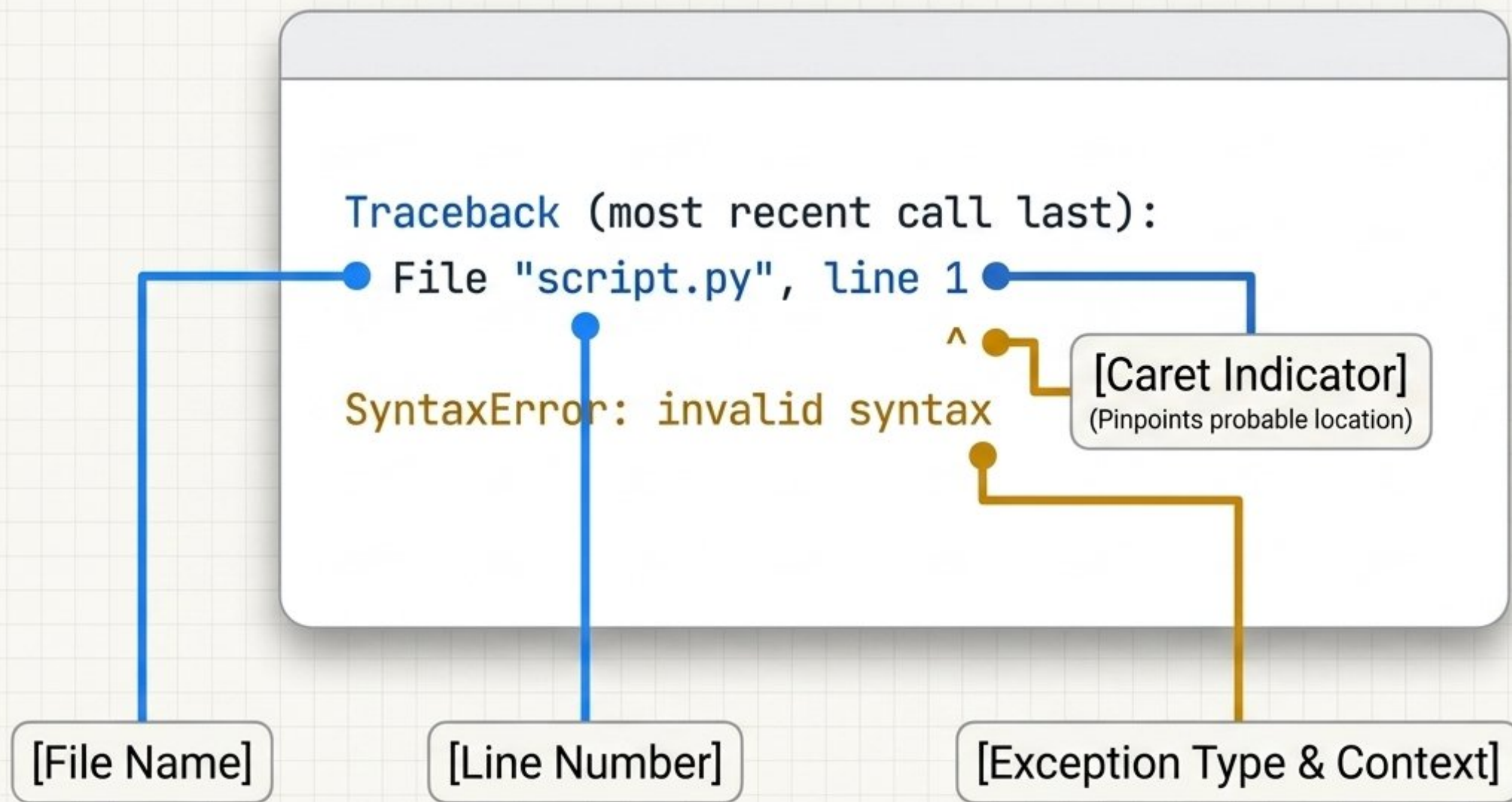
OOA groups related logic, preventing repetitive procedural code.
Everything in **Python** is an **object**.

Defensive Architecture: How Syntax Mitigates Failure



Advanced syntax is proactive debugging.

Anatomy of a Traceback



A traceback tells you exactly where things broke. Always read from the bottom up.

The Error Typology Matrix

Code Structure

SyntaxError
(missing brackets/colons)

IndentationError
(mixed tabs/spaces)

NameError
(undefined identifiers)

Data & Logic

TypeError
(string/int mismatch)

ValueError
(invalid int format)

IndexError
(out-of-range sequence)

KeyError
(missing dict key)

AttributeError
(missing object method)

ZeroDivisionError
(n/0)

Environment & System

ModuleNotFoundError
(missing pip install)

FileNotFoundError
(bad path)

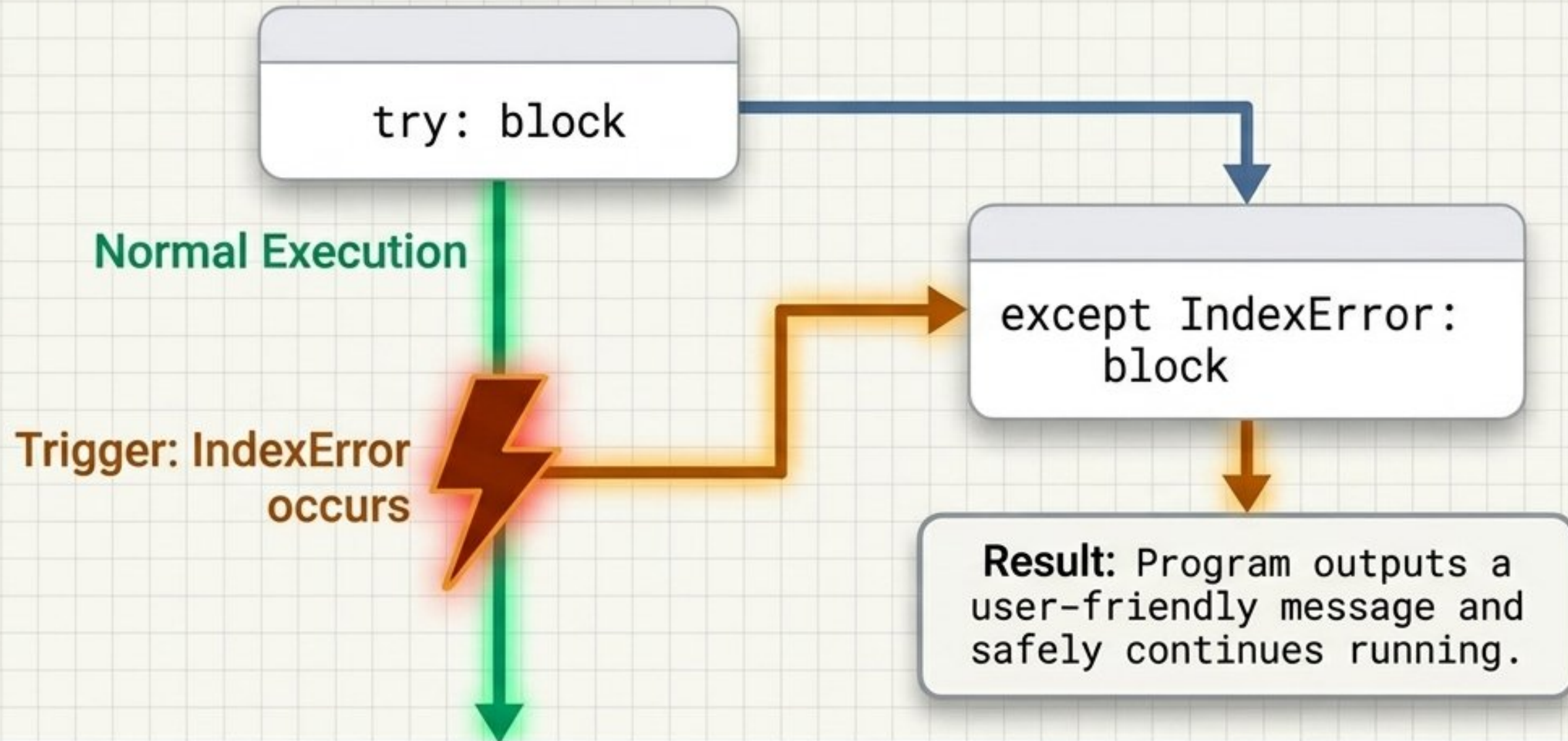
MemoryError
(system limit reached)

PermissionError
(lacking admin/sudo)

UnicodeError
(encoding conflict)

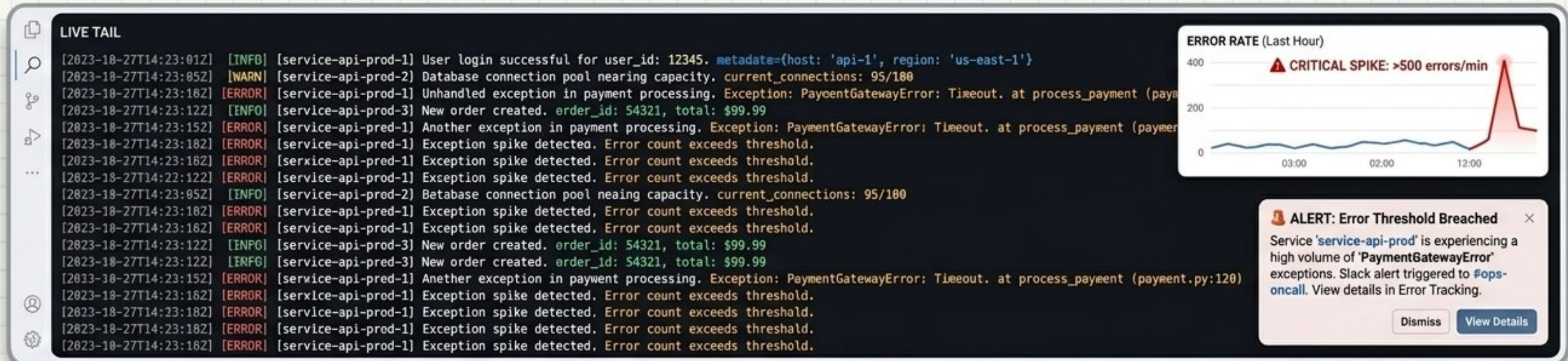
UnboundLocalError
(referencing local before assignment)

The Try/Except Circuit



Never let unpredictable user input crash your application.
Isolate volatile code blocks within specific exceptions.

The Watch: Production Log Centralization



The Collector

Ships logs reliably from servers/containers, enriching them with host and environment metadata.
Standardizes formats across all sources.

Live Tail

Follow incidents in real-time across all machines, bypassing the need to SSH into individual boxes.
Instant cross-fleet visibility.

AI-Native Error Tracking

Groups exceptions automatically, highlights frequent stack traces, and triggers Slack/Email alerts during error spikes.
Proactive issue identification.

Tracebacks tell you where it broke. Centralized logs tell you why it broke.